

RITE TALENT TRACKER

Technical & Functional Documentation

Architecture, Screen-by-Screen Walkthrough & Process Flow

Prepared for: Rite Software

Application: RITE Talent Tracker (Recruitment & Talent Pipeline Management)

Version: Working Draft — July 2026

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1. Introduction

This document is the combined technical and functional reference for RITE Talent Tracker. It complements the Business Object Document (data model, relationships, RBAC matrix) with an architecture overview, a screen-by-screen functional walkthrough with representative UI mockups, consolidated process flow charts, and a technical reference for developers (API routes, deployment, security).

Screen images in Section 3 are high-fidelity mockups built directly from the application's actual component structure and Redwood-navy design tokens (`src/styles.css`), reflecting real screen names, tabs, and fields — not screenshots of a live deployed instance.

- Audience: developers extending the application, QA/UAT reviewers, and functional stakeholders at Rite Software
- Companion document: RITE Talent Tracker — Business Object Document (data model & RBAC matrix)

2. System Architecture

The application follows a conventional four-tier architecture: a React single-page application, a single Express API layer, three supporting service modules, and a PostgreSQL data layer, all deployable via Docker Compose.

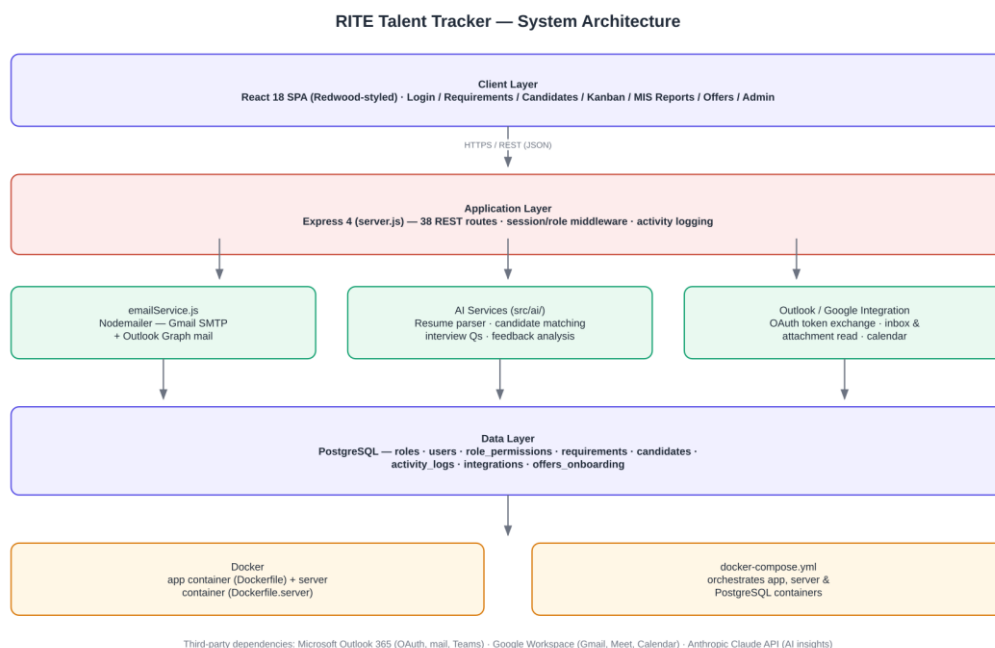


Figure 2.1 — System architecture: client, application, service, data and deployment layers

2.1 Technology Stack

Layer	Technology
Frontend	React 18 (Create React App), Context API, plain CSS design system (Redwood-navy palette)
Backend	Node.js, Express 4 — single server.js with 38 REST routes
Database	PostgreSQL — 7 core tables (see Business Object Document, Section 2)
Email	Nodemailer — Gmail SMTP and Outlook Graph mail
AI	Resume parsing, candidate matching, interview question generation, feedback analysis (src/ai/)
Auth	Local username/password + Outlook 365 OAuth
Containerization	Docker (Dockerfile, Dockerfile.server) + docker-compose.yml

3. Functional Documentation — Screen Walkthrough

Each subsection below pairs a representative screen mockup with an explanation of the screen's purpose, the business objects it reads/writes, and the roles that can access it (see the RBAC matrix in the Business Object Document, Section 5).

3.1 Login & Authentication

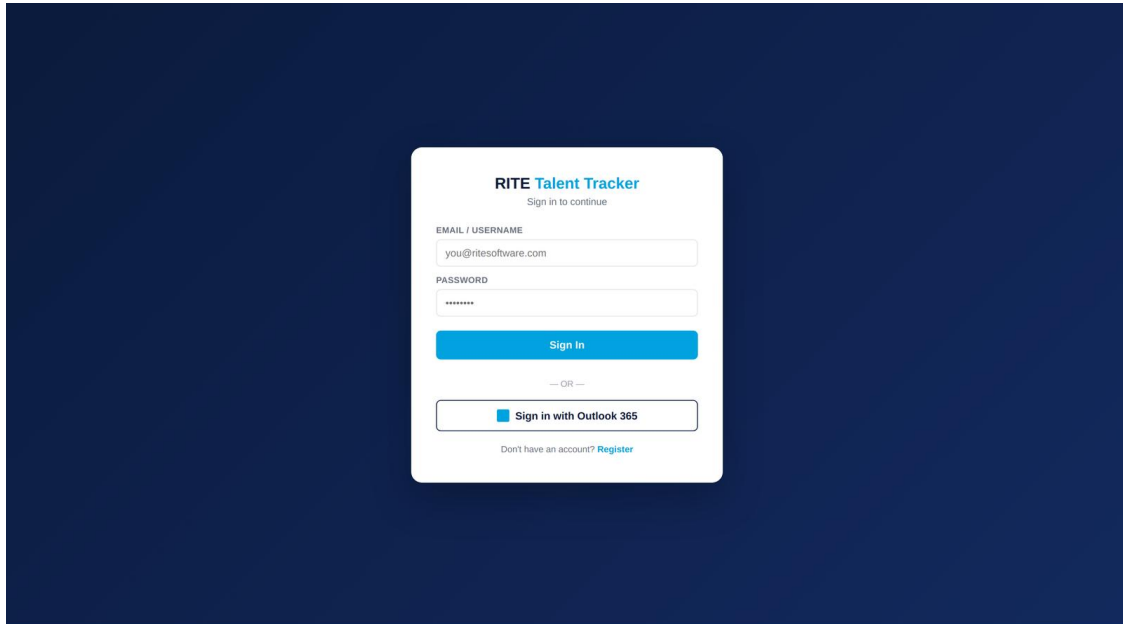


Figure 3.1 — Login screen (*src/components/Login.jsx*)

Supports two authentication paths: local username/password (validated against the User object) and Sign in with Outlook 365, which exchanges an OAuth token for an app session via `/api/auth/outlook-login`. A Register link is available for new local accounts, gated by an Admin approving the account's role afterward.

3.2 Requirements Dashboard

RITE Talent Tracker

Requirements

Candidates

Track Leads

Recruiter

MIS Reports

Offers & Docs

Admin

Requirements

47

Open Requirements

18

Staffing (Client)

29

Internal

6

Critical Priority

5

Stop-Gap Arrangements

Search by skill, client, owner...

All Verticals

All Clients

+ New Requirement

REQ ID	SKILL / SUB-SKILL	TYPE	CLIENT / VERTICAL	PRIORITY	STATUS	OWNER	NEED BY
RM00231	HCM > EBS	Staffing	Acme Financial Corp	Critical	Sourcing	AnilRaj	12-Jul-2026
RM00229	FIN	Internal	Rite Software	High	Open	GKM	18-Jul-2026
RM00227	Tech > React	Staffing	Nova Retail Group	Medium	Sourcing	Srinivas R	22-Jul-2026
RM00224	SCM	Staffing	Meridian Logistics	High	Partial	Seshendra	09-Jul-2026
RM00220	PPM	Internal	Rite Software	Low	Open	AnilRaj	30-Jul-2026

Figure 3.2 — Requirements list (src/components/Requirements.jsx)

The landing screen for Track Lead, Manager, DeliveryHead, Recruiter and Admin roles. KPI tiles summarize open, Staffing vs. Internal, Critical-priority and Stop-Gap requirements. The table is filterable by Vertical and Client (Section 6.3 of the Business Object Document) and each row shows skill/sub-skill, priority, and status badges.

3.3 New Requirement Form

RITE Talent Tracker

Requirements

Candidates

Track Leads

Recruiter

MIS Reports

Offers & Docs

Admin

New Requirement

Track Lead : Anil Raj

PN

New Requirement — Staffing Requirement

Core Info

Job Description & Hiring

Schedule & Management

Core Info

SKILL / SUB-SKILL

HCM > EBS

CLIENT

Acme Financial Corp

VERTICAL

Staffing

PRIORITY

Critical

TYPE

New Project

EXPERIENCE REQUIRED

6+ years

Job Description & Hiring

JOB DESCRIPTION

Oracle HCM Cloud – EBS integration specialist, payroll & absence management modules...

BILL RATE

\$68 / hr

TENURE OF THE PROJECT (CONTRACT POSITIONS)

6 months

STOP-GAP ARRANGEMENT

☐ No
 ☒ Yes — temporary cover

Schedule & Management

CLIENT HIRING MANAGER

Rebecca Shaw (Acme Financial Corp)

NEED BY DATE

12-Jul-2026

OWNER / MANAGER

AnilRaj / GKM

Cancel

Save Requirement

Figure 3.3 — New Requirement modal (src/components/RequirementModal.jsx)

A tabbed form — Core Info, Job Description & Hiring, Schedule & Management — reflecting the 7 July 2026 feedback: Priority now lives in Core Info for both Staffing and Internal requirements, the Job

Description is captured alongside Bill Rate, a Tenure of the Project field appears for contract positions, Stop-Gap is a first-class flag, and the Staffing Hiring Manager field is explicitly labelled Client Hiring Manager.

3.4 Candidate Pipeline — Kanban View

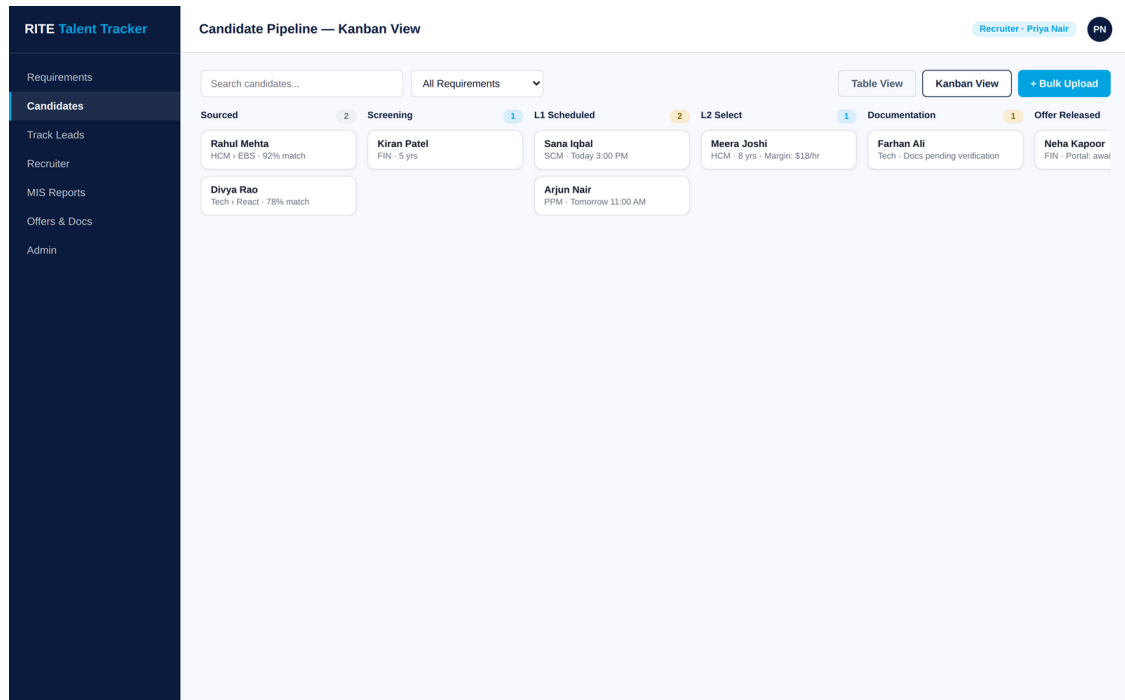


Figure 3.4 — Candidate Kanban board (src/components/PipelineKanban.jsx)

Gives recruiters a visual, per-candidate view of pipeline stage alongside the existing tabular Candidates view (src/components/Candidates.jsx), addressing the requested Kanban-per-candidate enhancement. Cards surface AI skill-match score, interview timing, or margin depending on stage. Bulk Upload is available directly from this screen.

3.5 Candidate Profile

RITE Talent Tracker

Requirements

Candidates

Track Leads

Recruiter

MIS Reports

Offers & Docs

Admin

Candidate Profile

Recruiter · Priya Nair

PN

Meera Joshi

CAND00412 · HCM · EBS · req RM00231

L2 Select

Profile

Interview Rounds

Feedback

Documents

CURRENT ROLE

Sr HCM Consultant

CURRENT COMPANY

Zenith Consulting

CURRENT LOCATION

Hyderabad, IN

WORK MODE PREFERENCE

Hybrid

EXPERIENCE (AI-FILLED)

8.0 yrs ⚡

EDUCATION (AI-FILLED)

M.Tech, IIT Kharagpur ⚡

CURRENT CTC

₹ 24,00,000 / yr

EXPECTED CTC

₹ 29,00,000 / yr

MONTHLY CTC (INR) ÷ USD/INR RATE

₹ 2,41,667 ÷ 90.0

SKILL MATCH SCORE (AI)

94% ⚡

MARGIN (BILL RATE - EXPECTED CTC)

\$18 / hr

CURRENT STAGE

L2 Select Hold

Figure 3.5 — Candidate profile modal (src/components/CandidateModal.jsx)

Fields marked ⚡ (Experience, Education, Skill Match Score) are AI-auto-filled from a bulk-uploaded resume and remain recruiter-editable. Current CTC and Expected CTC are tracked separately; Monthly CTC (INR) and the USD/INR rate stay interlinked. Margin is computed live as Bill Rate minus Expected CTC, and the Hold control is available directly from the profile header.

3.6 MIS Reports & Analytics

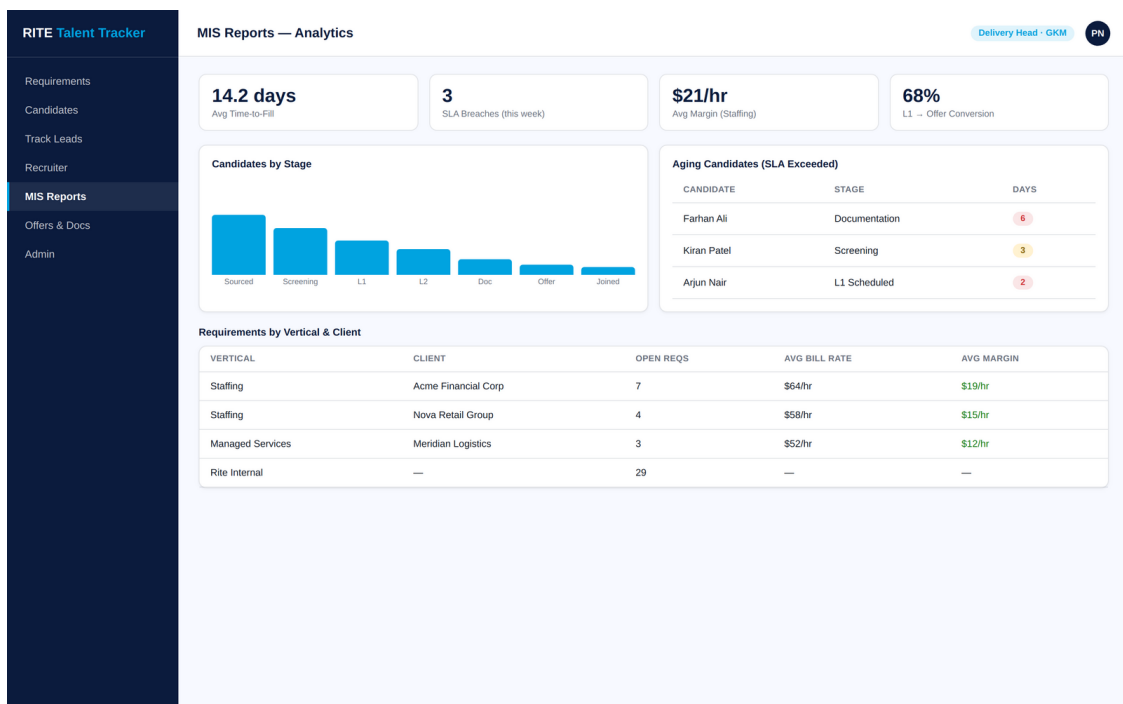


Figure 3.6 — MIS Reports (src/components/MISReports.jsx)

Aggregate reporting for Manager, DeliveryHead and Admin roles: time-to-fill, SLA breaches against the Stage SLA Rules table, average margin, pipeline funnel counts, and a per-Vertical / per-Client breakdown of open requirements, bill rate and margin.

3.7 Documentation Verification

The screenshot displays the 'Documentation Verification' screen in the RITE Talent Tracker application. The sidebar on the left contains navigation links: Requirements, Candidates, Track Leads, Recruiter, MIS Reports, **Offers & Docs**, and Admin. The main content area is titled 'Documentation Verification' and shows the candidate's name 'Farhan Ali (CAND00398)'. The form includes a 'CANDIDATE STAGE' dropdown set to 'Documentation Verification' and an 'OFFICIAL COMPANY EMAIL VERIFIED' section showing the email 'farhan.ali@newcompany.com' as 'Verified'. Below this is a table of 'Uploaded Documents' with columns for Document Type, File, Status, and Verified By. The table lists four documents: Government ID (Verified), Education Certificate (Verified), Experience Letter (Pending Review), and Latest Payslip (Rejected — unreadable). At the bottom right, there are two buttons: 'Request Re-submission' and 'Proceed to Offer Initiation'.

DOCUMENT TYPE	FILE	STATUS	VERIFIED BY
Government ID	farhan_id.pdf	Verified	Priya Nair
Education Certificate	farhan_mtech.pdf	Verified	Priya Nair
Experience Letter	farhan_exp_letter.pdf	Pending Review	—
Latest Payslip	farhan_payslip.pdf	Rejected — unreadable	Priya Nair

Figure 3.7 — Documentation Verification screen (new — see Business Object Document 6.24)

Sits between interview selection and offer initiation. Recruiters review uploaded identity, education and experience documents, cross-verify the candidate's official company email, and either request re-submission or proceed to offer initiation — writing a DocumentVerification record for the candidate.

3.8 Admin — Role Permissions Mapping

RITE Talent Tracker

Requirements

Candidates

Track Leads

Recruiter

MIS Reports

Offers & Docs

Admin

Admin — Access Control

Admin · System

PH

Users

Role Permissions

Integrations

Toggle checklist options to define which screens are visible to each system role. Login-credential access itself follows this same matrix.

ROLE	REQUIREMENTS	CANDIDATES	TRACK LEADS	RECRUITER	MIS REPORTS	OFFERS & DOCS	ADMIN
Admin	☒	☒	☒	☒	☒	☒	☒
DeliveryHead	☒	☒	☒	☒	☒	☒	☐
Manager	☒	☒	☒	☐	☒	☒	☐
TrackLead	☒	☒	☒	☐	☒	☐	☐
Recruiter	☒	☒	☐	☒	☒	☒	☐
Interviewer	☐	☒	☒	☐	☒	☐	☐

Reset to Default

Save Permissions

Figure 3.8 — Access Control screen (src/components/AccessControl.jsx)

Admin-only screen with three tabs — Users, Role Permissions, Integrations. The Role Permissions tab is the live editor for the RolePermissions object (Business Object Document, Section 2.6): toggling a checkbox immediately changes what that role sees, application-wide, without a redeploy. Admin's own row is always locked on. Per the 7 July feedback, access to viewing/resetting login credentials on the Users tab is itself gated by this same matrix.

4. Process Flow Charts

These flow charts are reproduced from the Business Object Document for convenience; see that document for the full field-level explanation behind each step.

4.1 End-to-End Recruitment Process Flow

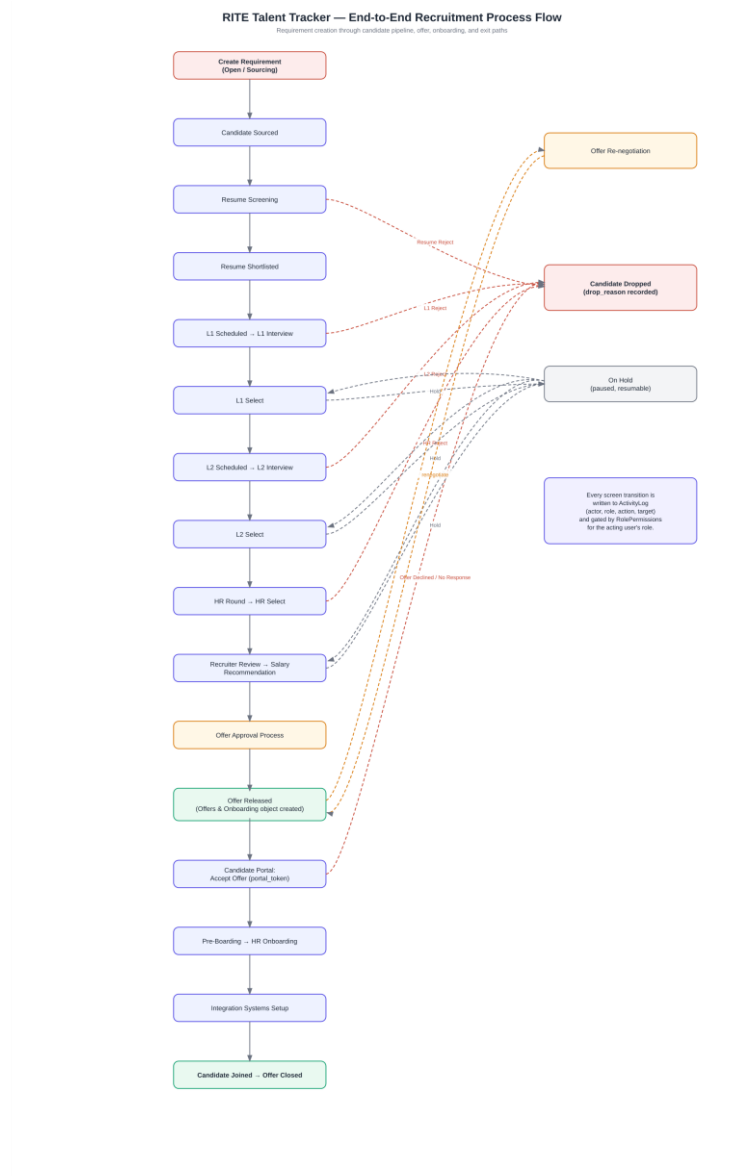


Figure 4.1 — Requirement creation through candidate pipeline, offer, onboarding and exit paths

4.2 Bulk Upload & AI Auto-Fill Flow

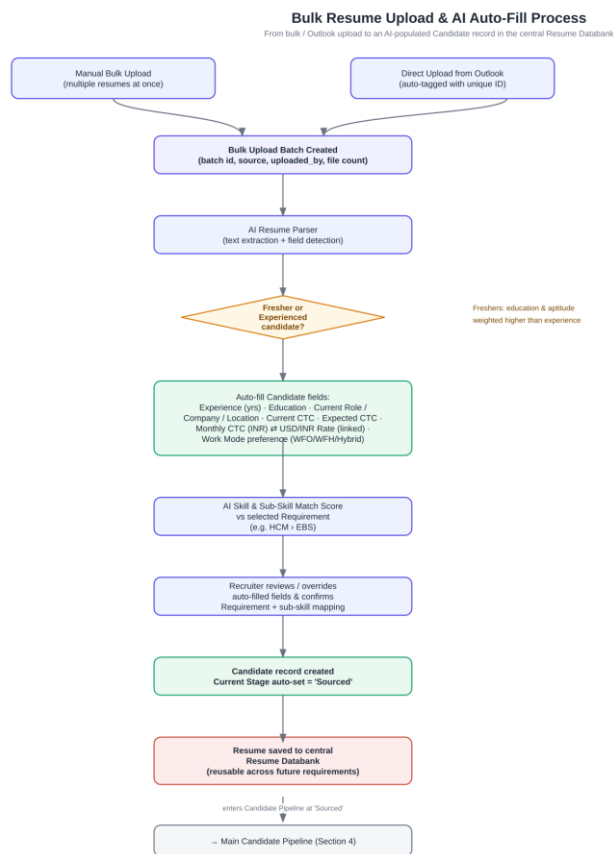


Figure 4.2 — From bulk / Outlook upload to an AI-populated Candidate record

4.3 Notification, Documentation & Margin Flow

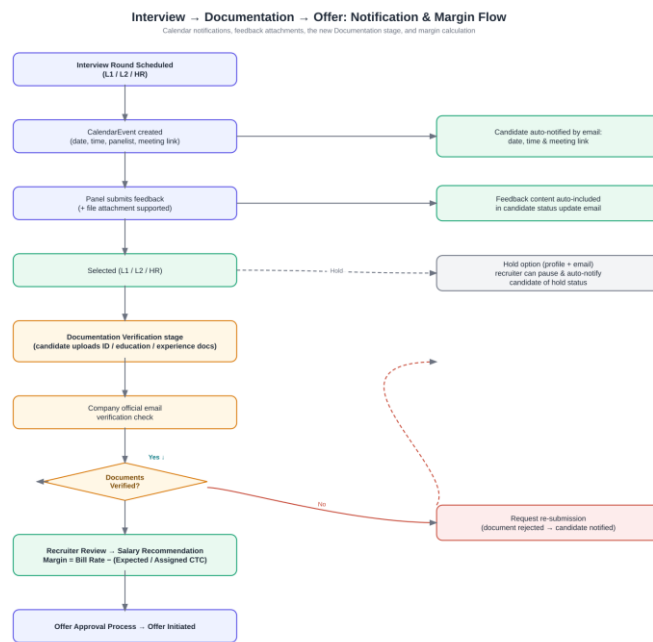


Figure 4.3 — Calendar notifications, Hold, feedback attachments, Documentation stage and margin calculation

4.4 Business Object Relationship Diagram

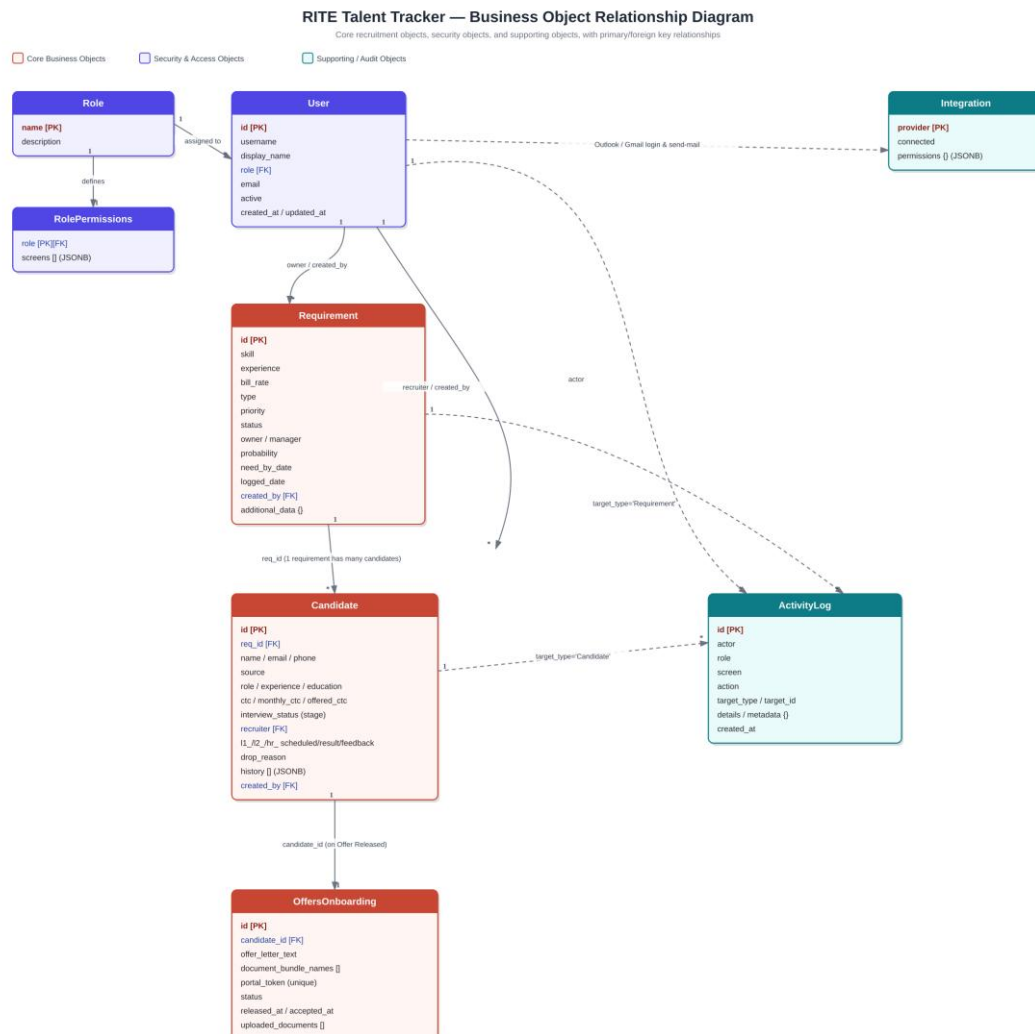


Figure 4.4 — Core, security and supporting business objects and their relationships

5. Technical Reference

5.1 REST API Reference

38 routes on the Express application layer, grouped by resource. Full request/response payloads are defined in `server.js`.

Auth

Method	Endpoint	Description
POST	/api/auth/login	Local username/password login
POST	/api/auth/register	Create a new local user account
POST	/api/auth/outlook-login	Exchange an Outlook OAuth token for an app session

Requirements

Method	Endpoint	Description
GET	/api/requirements	List all requirements
POST	/api/requirements	Create a requirement
PUT	/api/requirements/:id	Update a requirement
DELETE	/api/requirements/:id	Delete a requirement (cascades to candidates)

Candidates

Method	Endpoint	Description
GET	/api/candidates	List all candidates
POST	/api/candidates	Create a candidate
PUT	/api/candidates/:id	Update a candidate (stage, feedback, CTC, etc.)
POST	/api/generate-candidate-insights	AI-generated fit summary / insights for a candidate

Offers & Onboarding

Method	Endpoint	Description
GET	/api/offers	List all offer records
GET	/api/offers/candidate/:candidateId	Get the offer for a specific candidate
POST	/api/offers	Create/release an offer
PUT	/api/offers/:id	Update an offer record
POST	/api/offers/:id/reset	Reset an offer back to draft
GET	/api/offers/portal/:token	Candidate-facing portal — fetch offer by secure token
POST	/api/offers/portal/:token/accept	Candidate accepts the offer via the portal

Users & Roles

Method	Endpoint	Description
GET	/api/users	List all users
PUT	/api/users/:id/role	Change a user's role
PUT	/api/users/:id/status	Activate / deactivate a user
GET	/api/permissions	Get the RolePermissions matrix

Method	Endpoint	Description
POST	/api/permissions	Update the RolePermissions matrix

Activity & State

Method	Endpoint	Description
GET	/api/activity	Fetch the audit trail
POST	/api/activity	Write an audit trail entry
GET	/api/state	Bulk fetch of application state (used on app load)
GET	/api/health	Health check

Email

Method	Endpoint	Description
GET	/api/email-test	Send a test email to verify SMTP configuration
POST	/api/schedule-interview-email	Send an interview scheduling notification
POST	/api/cancel-interview-email	Send an interview cancellation notification
POST	/api/send-candidate-decision-email	Send a selection/rejection decision email

Outlook Integration

Method	Endpoint	Description
GET	/api/outlook/status	Check current Outlook connection status
POST	/api/outlook/token	Exchange an auth code for an Outlook access token
POST	/api/outlook/disconnect	Disconnect the Outlook integration
GET	/api/outlook/inbox	Read inbox messages
GET	/api/outlook/attachment/:messageId/:attachmentId	Download a specific email attachment

Integrations (org-level)

Method	Endpoint	Description
GET	/api/integrations	Get Microsoft / Google integration config
POST	/api/integrations	Update Microsoft / Google integration config

5.2 Database Schema

Full field-level schema, the entity-relationship diagram, and the candidate pipeline stage list are documented in the companion Business Object Document (Sections 2–4). In summary, the schema comprises 7 core tables — roles, users, role_permissions, requirements, candidates, activity_logs, integrations, offers_onboarding — plus the planned extensions (Client, Vertical, BulkUploadBatch, ResumeDatabank, CalendarEvent, DocumentVerification).

5.3 Security & Role-Based Access Control

- Authentication: local username/password, or Outlook 365 OAuth
- Authorization: every screen's visibility is governed by the database-backed RolePermissions object, editable live from Admin → Role Permissions Mapping
- Audit: every meaningful action across Requirement, Candidate, User and Admin screens is written to ActivityLog, polymorphically linked via target_type / target_id
- Candidate-facing surface: the Offer Portal uses a single-purpose, unguessable portal_token rather than a full account, scoped to one candidate's offer only
- Credential access: viewing or resetting another user's login credentials on the Admin → Users tab is itself gated by the RolePermissions matrix

5.4 Deployment

The application ships as three Docker services orchestrated by docker-compose.yml: the React app (Dockerfile), the Express API (Dockerfile.server), and PostgreSQL, seeded on first run from db/init.sql.

5.5 Third-Party Integrations

- Microsoft Outlook 365 — OAuth login, inbox & attachment read, calendar/Teams for interview scheduling
- Google Workspace — Gmail send, Google Meet, Calendar (integrations table, provider = 'google')
- Anthropic Claude API — AI-generated candidate insights, resume parsing, and interview question generation

6. Appendix

6.1 Environment Variables

Variable names only — actual values are environment-specific and are not reproduced here.

APP_URL	CORS_ORIGIN	PORT
REACT_APP_API_URL	PGHOST	PGPORT
PGDATABASE	PGUSER	PGPASSWORD
EMAIL_SERVICE	EMAIL_HOST	EMAIL_PORT
EMAIL_SECURE	EMAIL_USER	EMAIL_PASSWORD
EMAIL_REPLY_TO	EMAIL_CC	